



Point Pattern Analysis

Spatial Analysis VT25

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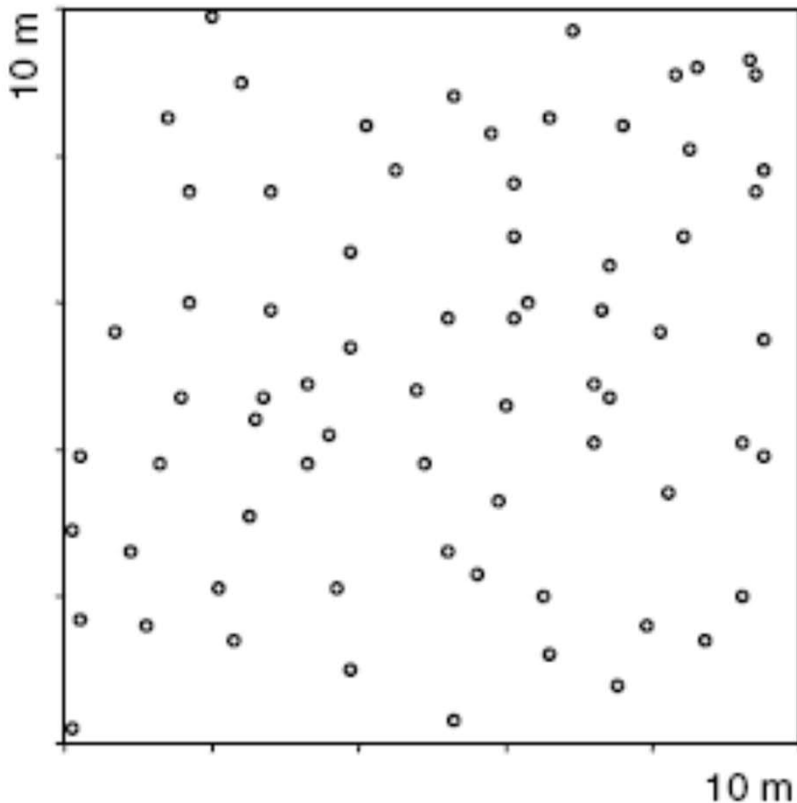
Format of Lecture

- What is a point pattern?
- What methods are available for identifying pattern in spatial point distributions (Quadrat Methods, Nearest Neighbour Index, G-function)?
- How can we formally test for the presence of pattern in spatial point distributions (simulation, statistical hypothesis testing)



What is a Point Pattern?

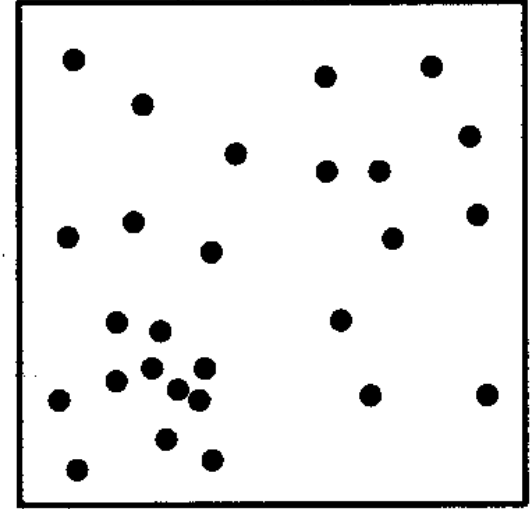
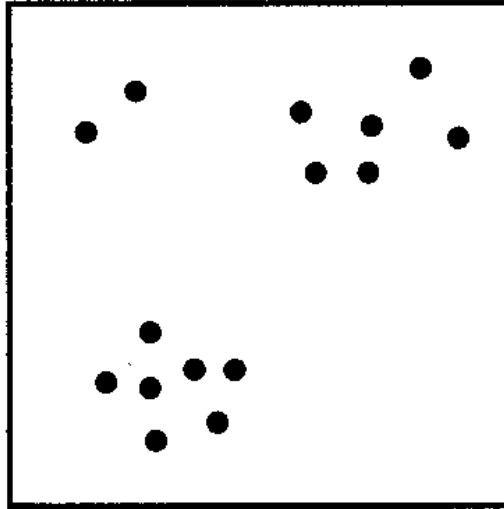
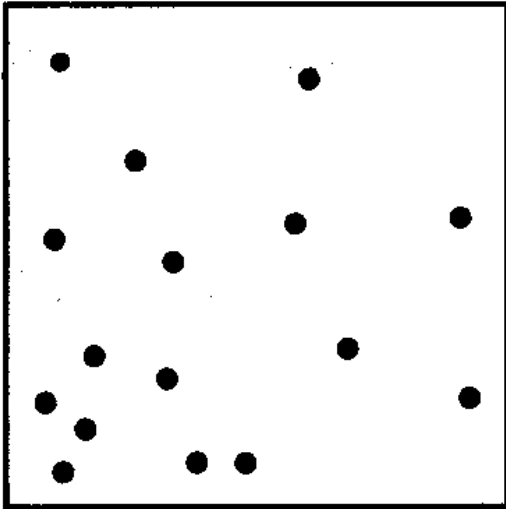
Swedish trees



- A set of points (**events**) distributed over a spatial domain
- Each **event** has a co-ordinate (x,y)
- There must be a complete census of **events** (not a sample)
- e.g. trees in a forest, locations of burglaries in Lund, 2011



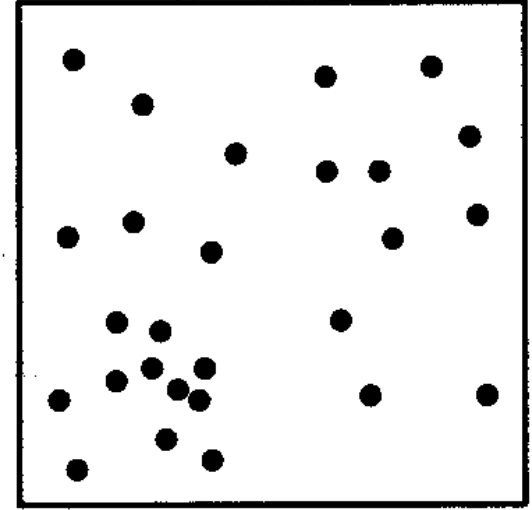
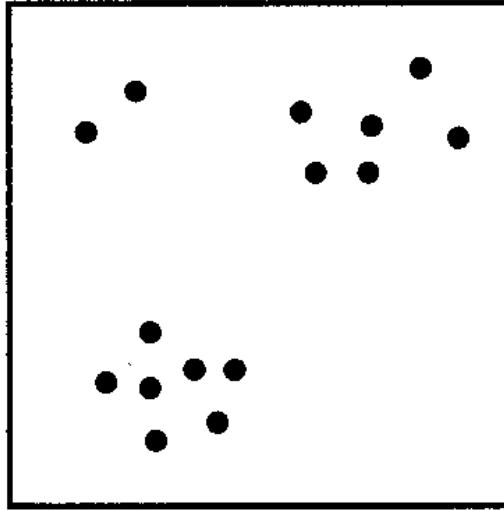
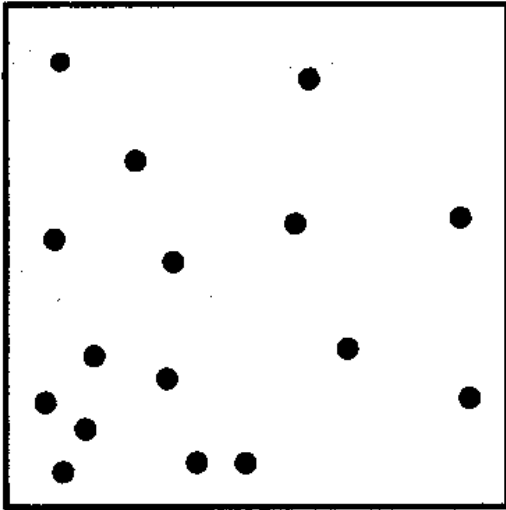
Questions to ask about Point Patterns



- Are the events random, clustered, or dispersed?
- Does the event intensity vary across the domain (area)?
- What processes give rise to these patterns (will not dwell on this too much in this lecture)?



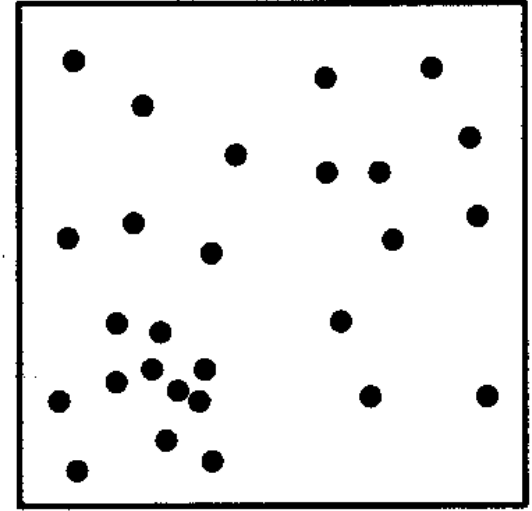
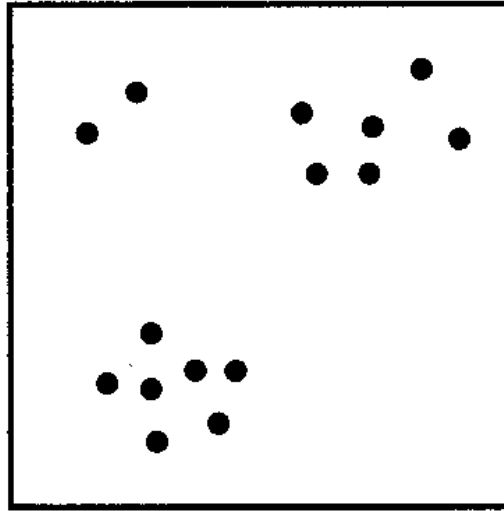
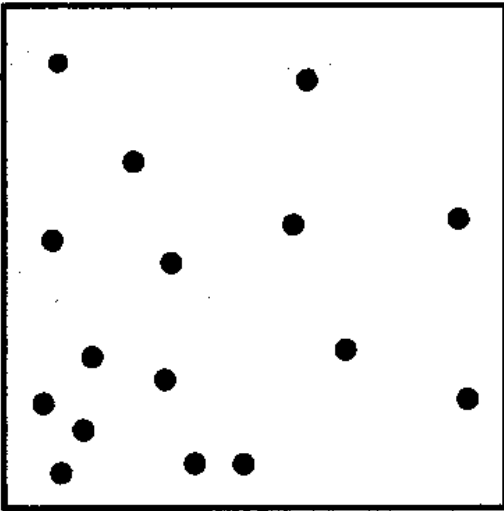
Testing for Pattern



- Visual (eyeballing maps, graphs)
- Statistical (testing the null hypothesis of complete spatial randomness)



Testing for Pattern

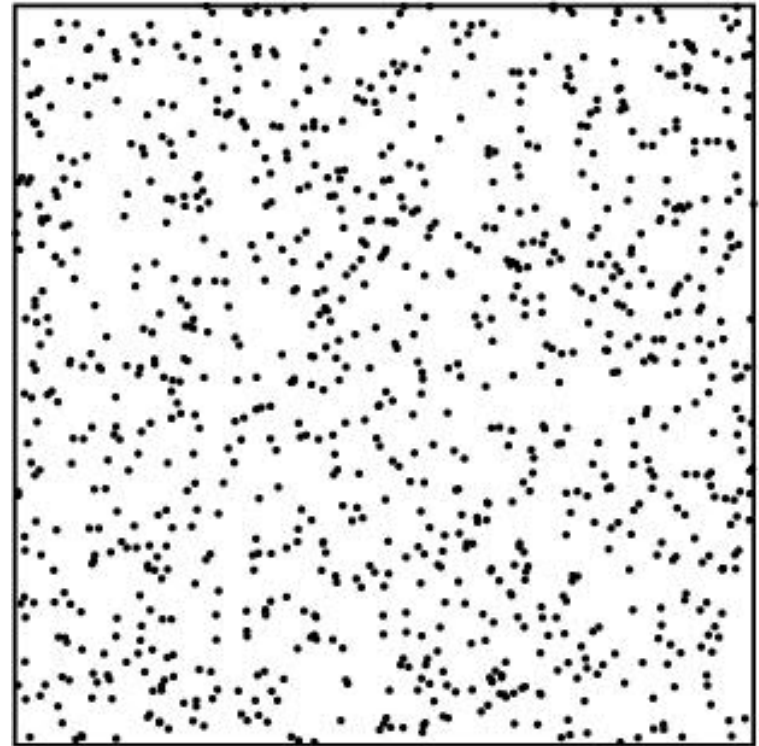


- Identifying clustering or dispersion in a point pattern requires setting up the null hypothesis of complete spatial randomness (CSR)
- The question posed: does the pattern conform to a Poisson distribution = null hypothesis?
- A test statistic is computed for the data and compared to the value of the statistic under CSR



Complete Spatial Randomness

- CSR is the simplest of all point patterns
- CSR means the intensity (no. events per unit area) of events does not vary, on average, across a region
- CSR also means that the location of an event is completely independent from the location of any other event

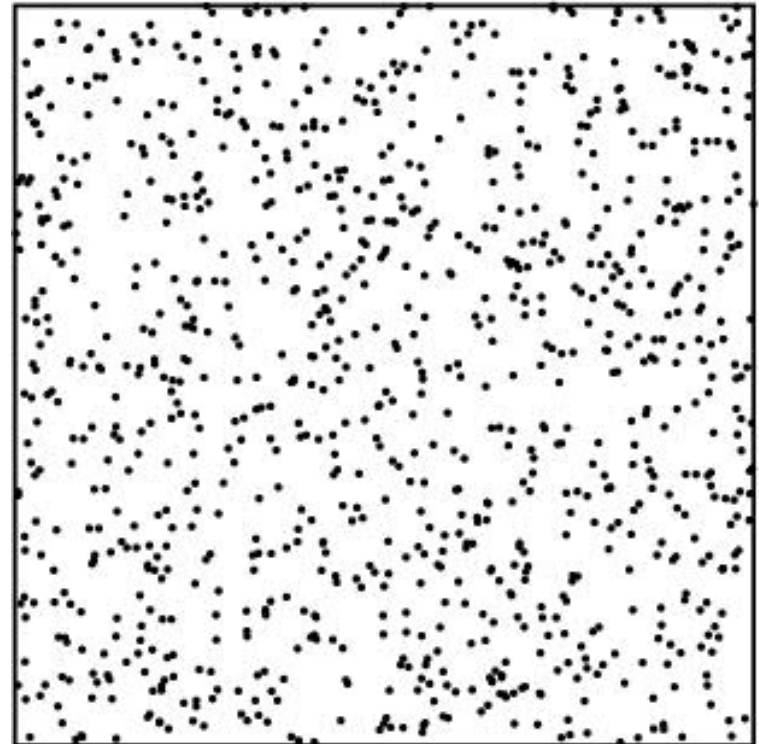


Spatial Poisson Process



Poisson Process

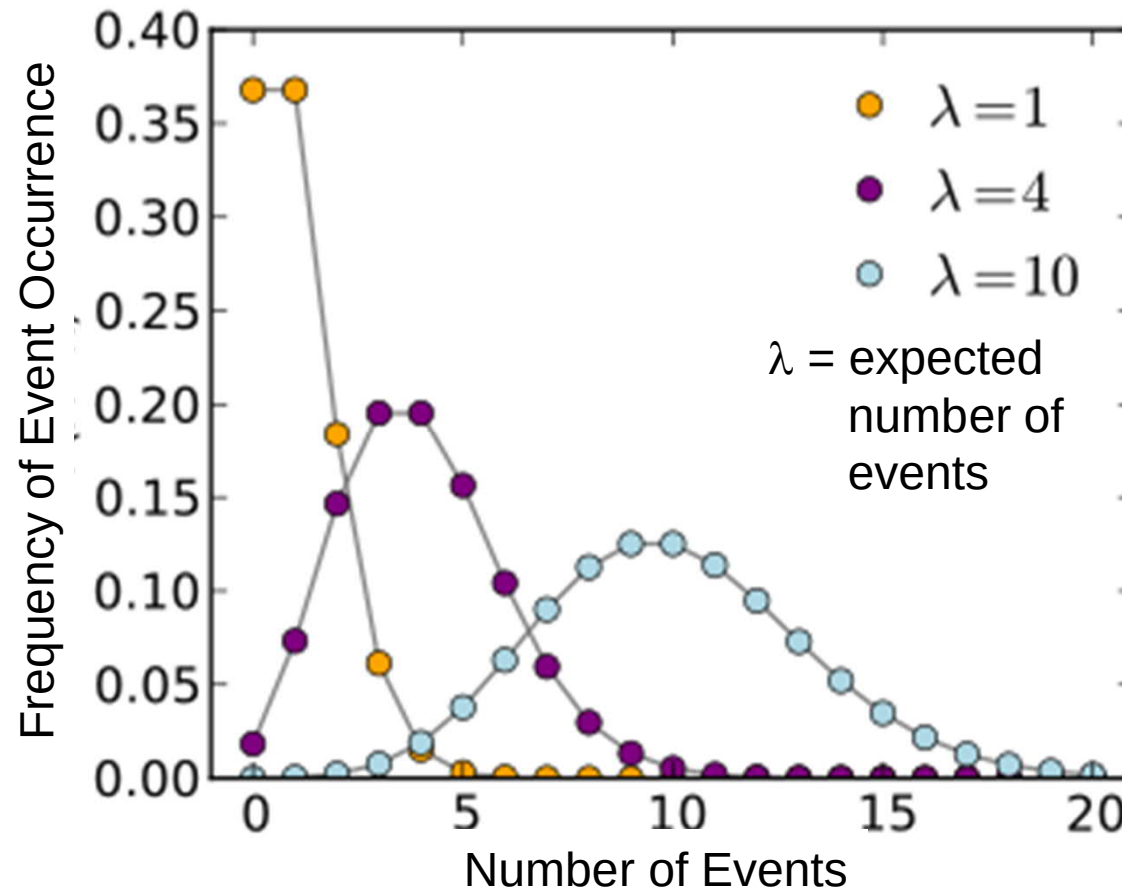
- Widely used statistical distribution for discrete events
- Used to describe pattern of random, point-like events in space (or time)
- If events occur randomly, and independently with a constant density in space (or rate in time), the count of these events per area conforms to a Poisson distribution
- Is often referred to as **Complete Spatial Randomness**



Spatial Poisson Process



Poisson Process



Probability of a given number of events occurring within a fixed space (time) interval. The Poisson distribution is discrete probability distribution.

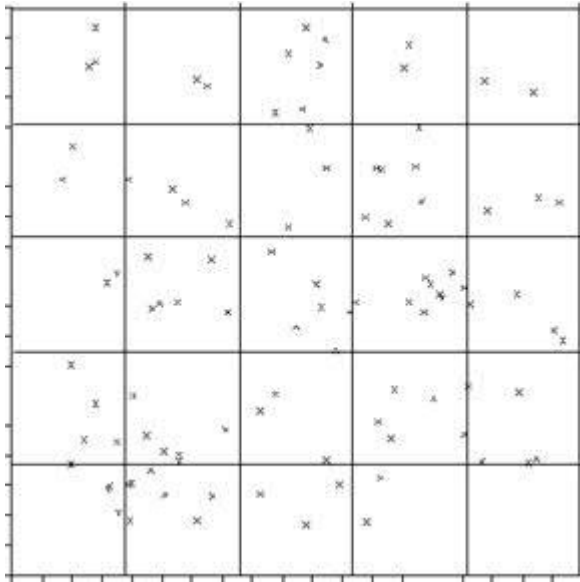


Tests for CSR

- Quadrat Method
- Nearest Neighbour Method
- G-Function



Quadrat Method



3	2	6	2	2
2	4	3	7	3
2	6	6	9	4
5	6	3	5	5
3	7	3	2	0

- Based on counting number of points occurring within a grid of a set quadrat size
- Frequency distribution is then developed for these points

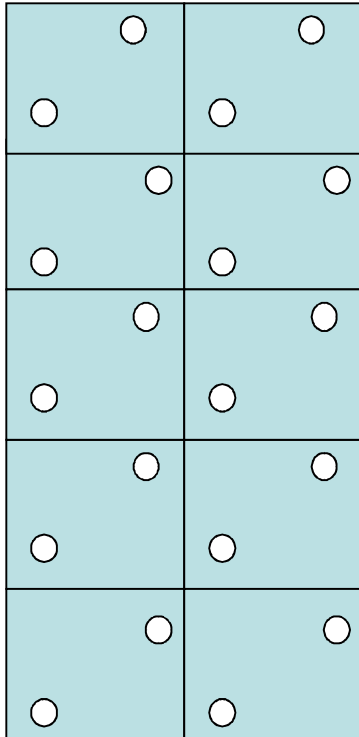


Quadrat

- Apply uniform grid over area
- count the number of points within each cell, to create the variable x
- Calculate variance and mean of x , and create the variance to mean ratio: $variance / mean = s^2 / \bar{x}$
- For a perfectly uniform (dispersed) distribution, $s^2 = 0$
 - Therefore, we expect a variance-mean ratio close to 0
- For a random distribution, $s^2 = \bar{x}$
 - Therefore, we expect a variance-mean ratio around 1
- For a clustered distribution, $s^2 > \bar{x}$
 - Therefore, we expect a variance-mean ratio above 1



Quadrat - Dispersed



2	2
2	2
2	2
2	2
2	2

Formula for s^2

$$\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{N - 1} = \frac{\sum_{i=1}^n X_i^2 - (\sum X)^2 / N}{N - 1}$$

Note:

N = number of Quadrats = 10

Ratio = $s^2 / \bar{X}$

Quadrat no.	X (no. points per quadrat)	X^2
1	2	4
2	2	4
3	2	4
4	2	4
5	2	4
6	2	4
7	2	4
8	2	4
9	2	4
10	2	4
	Sum=20	Sum=40

$$s^2 = 0.00$$

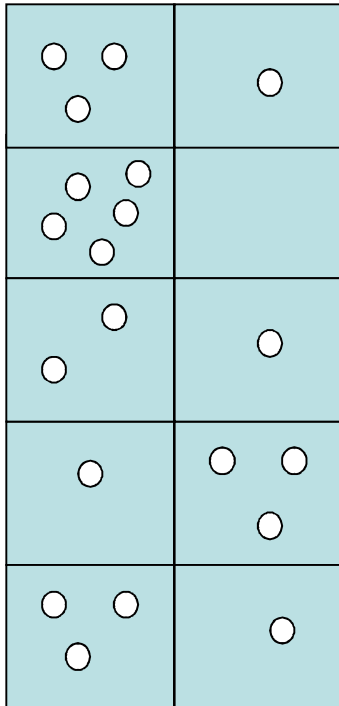
$$\bar{X} = 2.00$$

$$s^2 / \bar{X} = 0.00$$

UNIFORM/DISPERSED



Quadrat - Random



3	1
5	0
2	1
1	3
3	1

Formula for variance

$$\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{N - 1} = \frac{\sum_{i=1}^n X_i^2 - (\sum X)^2 / N}{N - 1}$$

Quadrat no.	X (no. points per quadrat)	X ²
1	3	9
2	1	1
3	5	25
4	0	0
5	2	4
6	1	1
7	1	1
8	3	9
9	3	9
10	1	1
	Sum=20	Sum=60

Note:

N = number of Quadrats = 10

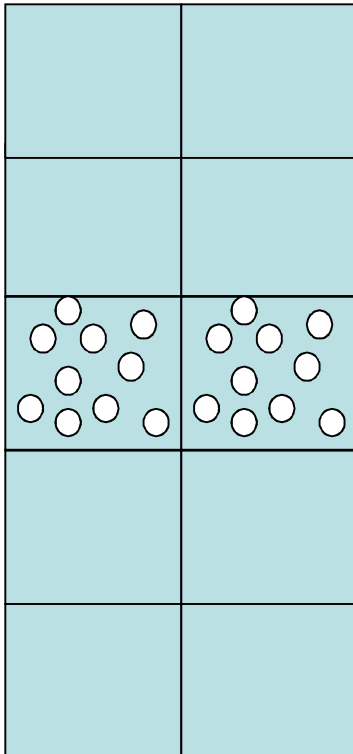
Ratio = $s^2 / \bar{X}$

$$\begin{aligned} s^2 &= 2.22 \\ \bar{X} &= 2.00 \\ s^2 / \bar{X} &= 1.11 \end{aligned}$$

RANDOM



Quadrat - Clustered



0	0
0	0
10	10
0	0
0	0

Formula for variance

$$\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{N - 1} = \frac{\sum_{i=1}^n X_i^2 - (\sum X)^2 / N}{N - 1}$$

Note:

N = number of Quadrats = 10

Ratio = $s^2 / \bar{X}$

Quadrat no.	X (no. points per quadrat)	X^2
1	0	0
2	0	0
3	0	0
4	0	0
5	10	100
6	10	100
7	0	0
8	0	0
9	0	0
10	0	0
	Sum=20	Sum=200

$$S^2 = 17.78$$

$$\bar{X} = 2.00$$

$$s^2 / \bar{X} = 8.89$$

CLUSTERED



Hypothesis Testing

- H_0 = no pattern (random distribution)
- H_1 = pattern exists (clustered or dispersed)
- How to we go about unambiguously testing this hypothesis?



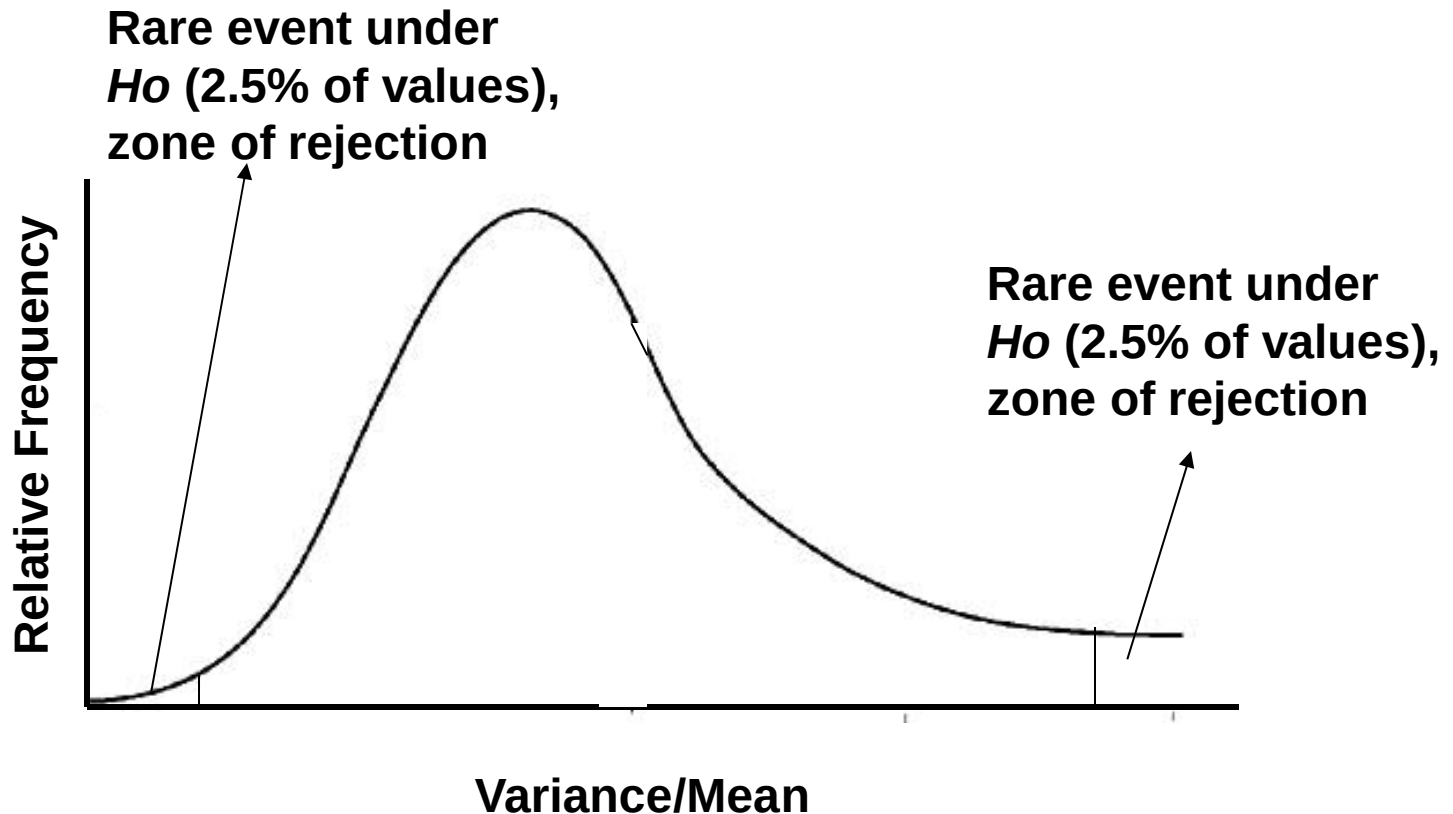
1. Derive the sampling distribution via simulation

- The benchmark to which we compare our actual distribution is a random distribution: we must simulate the H_0 *a large number of times*
- We therefore randomly distribute points across the region, not once, but > 1000 times, computing $s^2 / \bar{x}$ every time – in order to derive the sampling distribution of $s^2 / \bar{x}$
- We draw up the histogram of the results



1. Derive the sampling distribution via simulation

Sampling distribution of variance/mean when $H_0 = \text{true}$



If the variance/mean of our observations fall beyond the critical regions, then we reject H_0 of random pattern: our likelihood of rejecting H_0 when it is actually true) is 5% ($p = 0.05$)



2. Use Chi Square Test

- A significance test can be conducted based upon the chi-square test statistic, because $(s^2/\bar{x})(N-1)$ follows a chi-square distribution (for $N < 30$)
- The test statistic is given by:

$$= \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\bar{X}} = \frac{\sum_{i=1}^n X_i^2 - [(\sum X)^2 / N]}{\bar{X}}$$

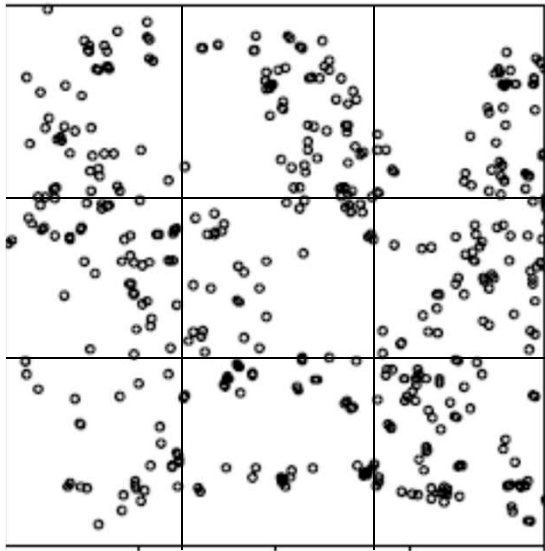
- The test will ascertain if a pattern is significantly different from CSR
- The values of the test statistics in our cases would be:

<u>random</u>	<u>dispersed</u>	<u>clustered</u>
$\frac{60 - (20^2)/10}{2} = 10$	$\frac{40 - (20^2)/10}{2} = 0$	$\frac{200 - (20^2)/10}{2} = 80$

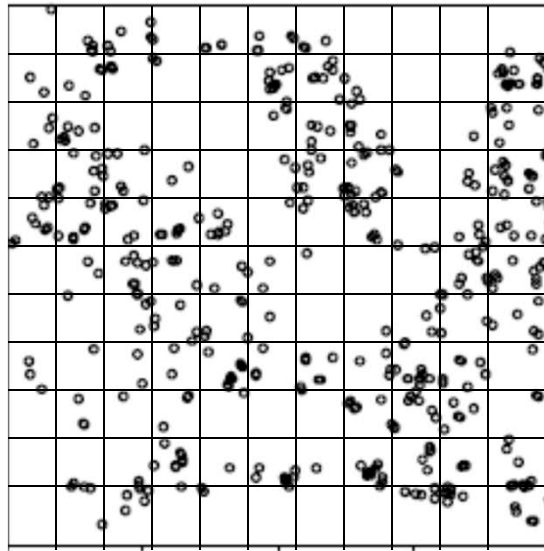
- For degrees of freedom: $N - 1 = 10 - 1 = 9$, the critical values at the 5% level ($p=0.025$, $p=0.975$) are 19.023 and 2.700
- Thus, there is only a 5% chance of obtaining a value greater than 19.023, or less than 2.700 if the points had been allocated randomly. Since our test statistic for the clustered (uniform) pattern is 80 (0), we conclude that there is less than a 2.5% chance that the clustered (dispersed) pattern could have resulted from a random process



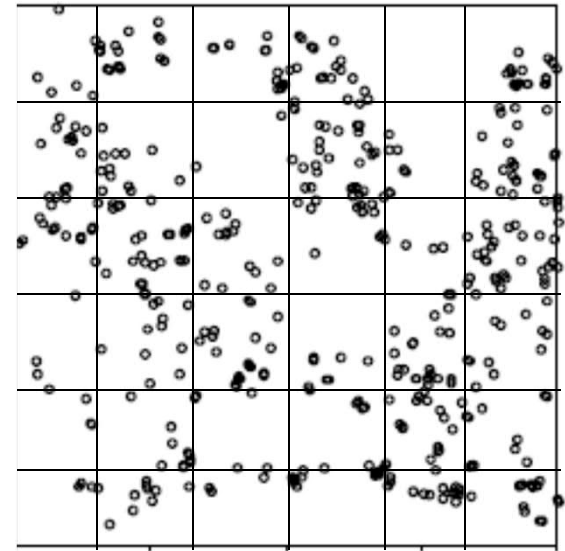
Problem Selecting Quadrat Size



Too large – too many quadrats with similar # of points



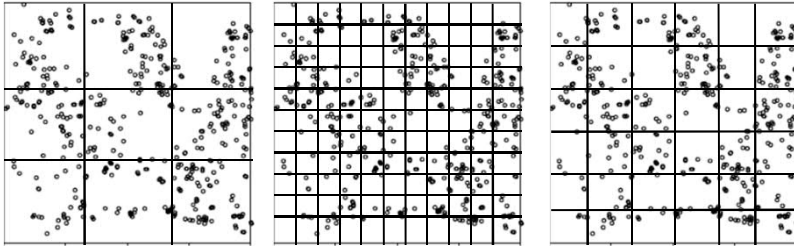
Too small – too many quadrats with 0 points



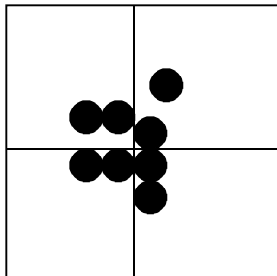
Just right – provides range in # of points per quadrat



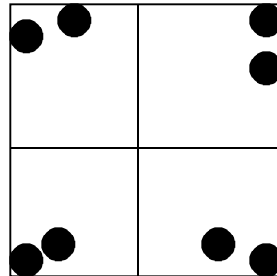
Issues with Quadrat Method



Must test effect of
quadrat size on result

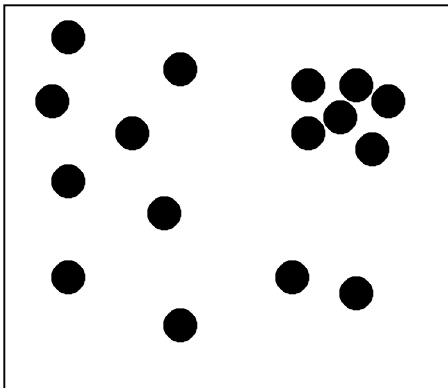


These two
arrangements
will give the
same answer



Based on the density of
points, not their
arrangement relative to
one another- not a true
measure of pattern

Overall
dispersion
with local
clustering



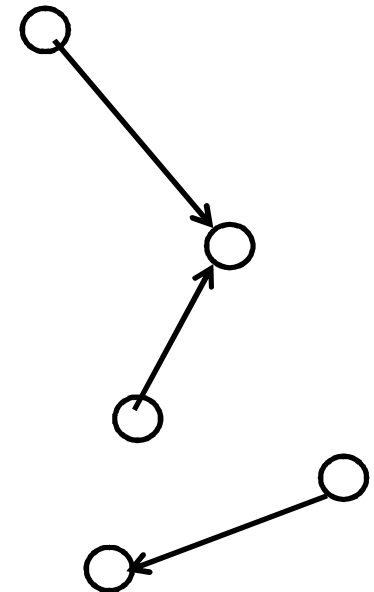
Is a global measure with
one value, variations w/i
regions not recognized



Nearest Neighbour Index

- Uses distances between points
- It compares:
 - the mean of the distance observed between each point and its **nearest** neighbour
 - with the expected mean distance if the distribution was random:

$$R = \frac{R_o \text{ (Observed Mean Distance)}}{R_e \text{ (Expected Mean Distance)}}$$





Nearest Neighbour Index

Nearest Neighbour Index

$$R = \frac{R_0}{R_e} = \frac{\bar{x}}{1/(2\sqrt{\lambda})} = \frac{\text{observed mean distance}}{\text{expected mean distance}}$$

where: λ = number of points per unit area

- for a random pattern, $R = 1$
- for clustered pattern, $R = 0$
- for dispersed pattern, $R = 2.149$



Nearest Neighbour – Significance Testing

- If many random patterns are examined, the variance of the mean distances between the neighbours is:

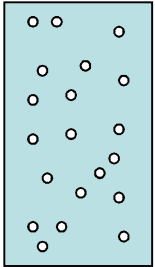
$$V[R_e] = \frac{4 - \pi}{4\pi\lambda n}$$

where n = number of points

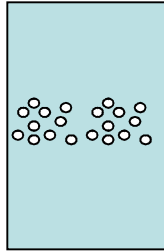
- A Z-test can be used to test H_0 :

$$z = \frac{(R_0 - R_e)}{\sqrt{V[R_e]}} = \frac{R_0 - R_e}{\sqrt{(4 - \pi)/(4\pi\lambda n)}} = 3.826(R_0 - R_e)\sqrt{\lambda n}$$

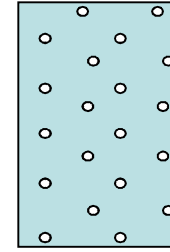
- Z has a normal frequency distribution with mean = 0 and variance = 1 and if Z is below -1.96 or above +1.96, we are “95% confident that the distribution is not randomly distributed”



RANDOM



CLUSTERED



UNIFORM

Point	Nearest Neighbour	Distance
1	2	1
2	3	0.1
3	2	0.1
4	5	1
5	4	1
6	5	2
7	6	2.7
8	10	1
9	10	1
10	9	1
		10.9

$$R_o = 1.09$$

$$\lambda = 10/50$$

$$R_e = 1.12$$

$$R = 0.98$$

$$Z = -0.15$$

Point	Nearest Neighbour	Distance
1	2	0.1
2	3	0.1
3	2	0.1
4	5	0.1
5	4	0.1
6	5	0.1
7	6	0.1
8	9	0.1
9	10	0.1
10	9	0.1
		1

$$R_o = 0.1$$

$$\lambda = 10/50$$

$$R_e = 1.12$$

$$R = 0.09$$

$$Z = -5.51$$

Point	Nearest Neighbour	Distance
1	3	2.2
2	4	2.2
3	4	2.2
4	5	2.2
5	7	2.2
6	7	2.2
7	8	2.2
8	9	2.2
9	10	2.2
10	9	2.2
		22

$$R_o = 2.2$$

$$\lambda = 10/50$$

$$R_e = 1.12$$

$$R = 1.97$$

$$Z = 5.86$$

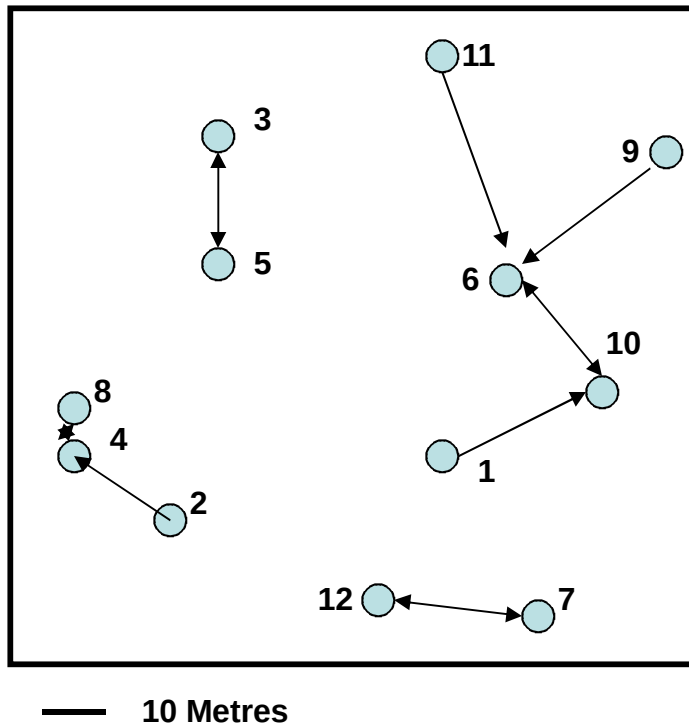


Nearest Neighbour - Issues

- Advantages
 - Unlike quadrats, the NNI considers distances between points
 - No quadrat size problem
- BUT
 - Very dependent on the size and shape of the study region
 - Long narrow regions necessarily have low distances because of shape
 - Edge effects - near points could occur just outside study area
 - Clustering is only detected on a fine spatial scale since only nearest neighbour is considered but not 2nd or 3rd order neighbours
 - Does not incorporate 'background factors' e.g. does clustering exist independent of e.g. population factors, e.g. due to underlying trends?



G Function



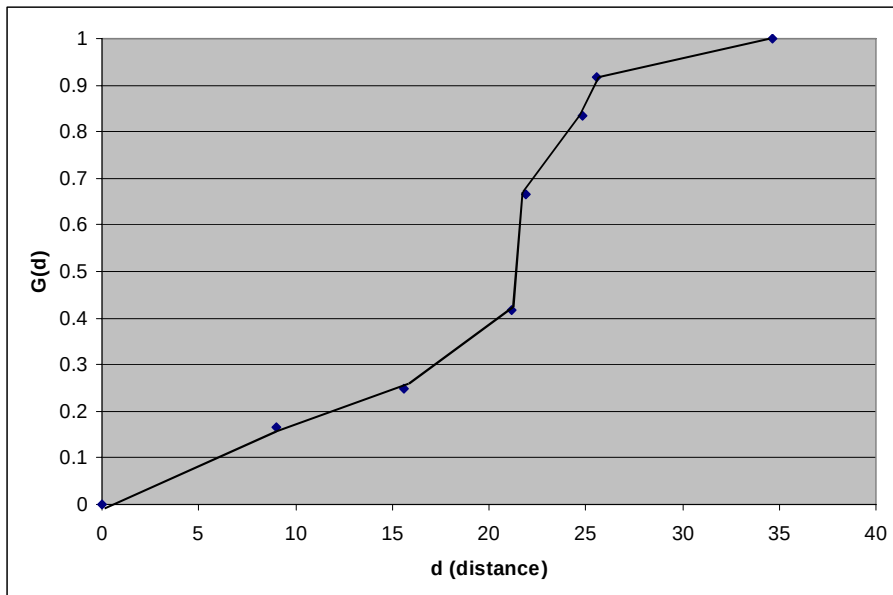
Event	X	Y	NN	d _{min}
1	66.22	32.54	10	25.59
2	22.52	22.39	4	15.64
3	31.01	81.21	5	21.14
4	9.47	31.02	8	9.00
5	30.78	60.10	3	21.14
6	75.21	58.93	10	21.94
7	79.26	7.68	12	24.81
8	8.23	39.93	4	9.00
9	98.73	42.53	6	21.94
10	89.78	42.53	6	21.94
11	65.19	92.08	6	34.63
12	54.46	8.48	7	24.81

$$G(d) = \frac{\text{no. point pairs where } d_{\min} \leq d}{\text{no. points in study area}} = \text{fraction of all NN distances less than or equal to } d$$



Interpreting G Function Plot

**$G(d)$ = fraction of all NN distances
less than or equal to d**



Clustered – G increases rapidly at small distances

**Dispersed – G increases slowly up to distances where
most events are spaced, then increases
rapidly**

Event	X	Y	NN	d _{min}
1	66.22	32.54	10	25.59
2	22.52	22.39	4	15.64
3	31.01	81.21	5	21.14
4	9.47	31.02	8	9.00
5	30.78	60.10	3	21.14
6	75.21	58.93	10	21.94
7	79.26	7.68	12	24.81
8	8.23	39.93	4	9.00
9	98.73	42.53	6	21.94
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12	54.46	8.48	7	24.81



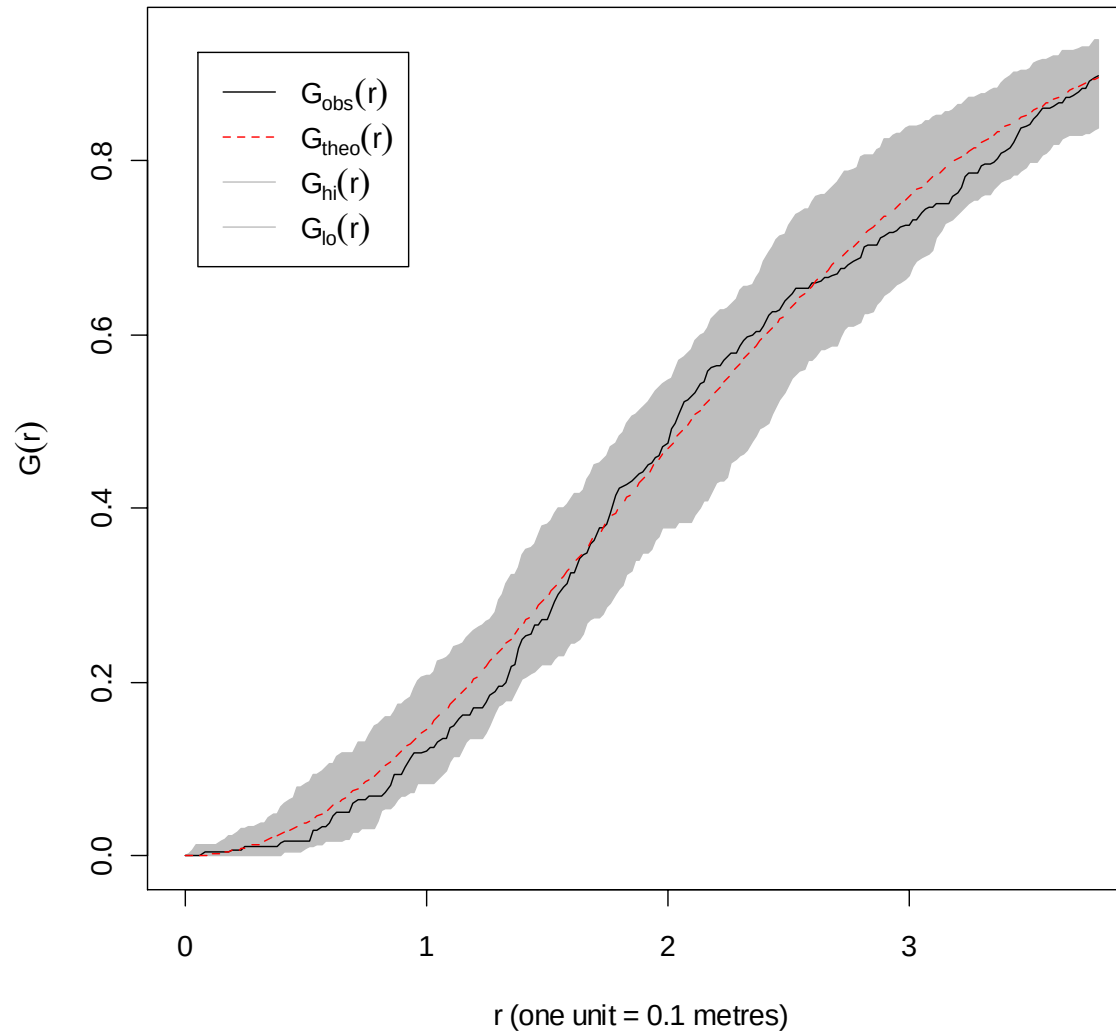
Significance Test of G Function

- Q: is there a significant departure from CSR?
- Create a large number (hundreds, thousands) of random fields with same number of events (produces a reference distribution)
- Produce G function for each random field
- Rank all the points for each d for all simulations
- Extract upper and lower 5th and 95th percentile $G(d)$ values for each d
- Plot them as 95% confidence intervals
- Does the empirical function fall within the envelope?



Envelope in R - Lab

`envelope(Z, Gest, nsim = 200)`





- Read: Rogerson, P.A. (2001) Statistical Methods for Geography, Sage, London. pp. 154-163